

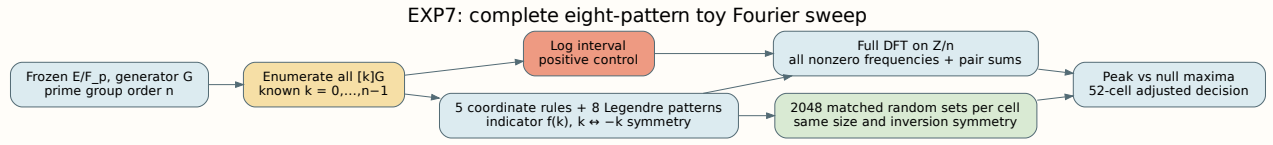
EXP7: complete Legendre-pattern Fourier sweep

Measured 2026-10-08 UTC. The 20261007 study-directory label is corrected in the [date erratum](#). This is the completed first item of the proposed six-item list, on four **toy** prime-order curves. No candidate met the declared adjusted lead rule. Items 2-6 were not run.

Protocol, sources, and population

The [original protocol](#) and its [precommitted all-pattern extension](#) define the membership rules, nulls, controls, and decisions. The original [512-null run](#) tested only the +++ Legendre pattern and is preserved with its [pilot report](#). The final [2048-null run](#) tests all eight sign patterns of the Legendre symbols of x , $x+1$, $x+2$, alongside B-smooth x , bounded continued-fraction quotients of x/p , base-3 Cantor digits, bounded base-2 Hamming weight, bounded Farey height, SHA-256 of x , one random inversion-pair set, and a log interval. Zero Legendre values belong to none of the eight sign-pattern cells. The [implementation](#) at measured revision `d3c26e537cbbd2e081f38631d4cd1784e03e407c` was committed before the run. The [compact CSV](#) and quantitative figure are derived from the complete JSON.

For $P=[k]G$, the indicator $f(k)$ selects points satisfying one predicate. The arbitrary-length FFT computes all nonzero character coefficients $F(j)=\sum_k f(k) \exp(-2\pi ijk/n)$ and records the maximum normalized by $\sqrt{m(1-m/n)}$, where m is the set size. Each coordinate case is compared with 2048 uniformly sampled, exact-size random sets of inversion pairs $\{k, -k\}$. This matches the unavoidable $x(P)=x(-P)$ symmetry. The log interval is compared with unrestricted exact-size random sets. The null statistic is the **maximum across all nonzero frequencies**, not a prespecified frequency. The program also inverse-transforms $F(j)^2$ to check ordered-pair sum counts and saves every membership bitset for exact spectral replay.



Editable [Graphviz source](#) and [vector PDF](#) accompany the diagram.

Bits	p	a	b	Prime n	G=(x,y)
8	179	51	41	191	(108,39)
10	739	242	567	719	(559,630)
12	2749	1936	200	2749	(1388,830)
14	10177	3690	7829	10321	(3481,6440)

p	Exact ICV1
179	ICV1:fp-179:-11:191:96:unk:unk:r:a52b529ce9ce
739	ICV1:fp-739:21:719:614:unk:unk:r:a47a74d0df47
2749	ICV1:fp-2749:1:2749:2173:unk:unk:r:096cb4f9e068
10177	ICV1:fp-10177:-143:10321:5678:unk:unk:r:b65f5c0e2b63

Each curve has cofactor 1 and $[n]G=0$. Exact enumeration checked n distinct on-curve points and counted exactly n group additions, zero doublings, and zero scalar multiplications per curve. The complete points in log order and $f(k)$ bitsets are in the JSON. This diagnostic starts with every toy log k known; it is not a cheap membership oracle for an unknown log. Curve search cost is outside the enumeration count.

Observations and declared decision

EXP7 all-pattern Fourier peaks against matched nulls

Cell = observed maximum / 95th percentile of 2048 same-size null maxima

	p=179 n=191	p=739 n=719	p=2749 n=2749	p=10177 n=10321
B-smooth x	0.97×	0.81×	0.83×	0.87×
CF quotients ≤16	0.86×	0.91×	0.77×	0.79×
Cantor digits	0.80×	0.77×	0.87×	0.95×
Hamming weight	0.97×	0.89×	0.90×	0.85×
Farey height	0.77×	0.87×	0.77×	0.79×
Legendre +++	0.89×	1.06×	0.86×	0.82×
Legendre ++-	0.96×	0.72×	0.86×	0.81×
Legendre +-+	1.01×	0.96×	0.73×	0.86×
Legendre +--	0.84×	0.80×	0.96×	0.84×
Legendre --+	0.75×	0.72×	0.91×	0.77×
Legendre -+-	0.91×	0.85×	0.78×	0.81×
Legendre ---+	1.04×	0.92×	0.77×	0.78×
Legendre ---	0.69×	0.78×	0.90×	0.82×
SHA(x) control	0.74×	0.68×	0.85×	0.79×
Random-pair control	0.79×	0.68×	0.84×	0.78×
Log-interval control	2.60×	4.70×	8.55×	15.53×

≤ null 95th percentile
 > null 95th percentile
 > 2× null 95th percentile

52 candidate cells; threshold $p \leq 0.05/52$. A colored cell alone is not a lead.

The chart has editable [Rust source](#), a [Cargo wrapper](#), [CSV data](#), and a [vector PDF](#). Each ratio is the observed normalized maximum divided by the 95th percentile of its 2048 null maxima. Yellow marks a ratio above 1, which alone is **not** the adjusted lead rule.

The predeclared family has 13 coordinate predicates × 4 curves = **52 candidate cells**, so a lead needs empirical $p \leq 0.05/52 = 0.000961538$. The minimum possible empirical $p=1/2049=0.000488043$; only zero null exceedances can cross the threshold. The JSON reports **zero** adjusted leads. The smallest candidate empirical value is the +++ Legendre cell on $p=739$, $n=719$: $m=84$, observed $Z=4.005158$, null 95th percentile $Z=3.761062$, ratio 1.064901 , and **31** of 2048 null maxima at least as large. Thus $p=(31+1)/2049=0.0156174$, and its Bonferroni adjusted value is 0.812103 . It remains the largest can-

didate ratio and smallest candidate p, without meeting the lead rule.

Candidate	Minimum empirical p across curves	Maximum peak/null-95 ratio
B-smooth x	0.07028	0.969
Continued fraction ≤ 16	0.16057	0.912
Base-3 Cantor digits	0.11713	0.954
Base-2 Hamming weight	0.07565	0.972
Farey height	0.29575	0.871
Legendre +++	0.01562	1.065
Legendre ++-	0.08541	0.964
Legendre +-+	0.04783	1.007
Legendre +--	0.10249	0.962
Legendre --+	0.22596	0.913
Legendre -+-	0.15471	0.914
Legendre ---	0.02879	1.041
Legendre ---	0.26696	0.898

The three candidate cells above their own null 95th percentiles are +++ at $p=739$ ($p=0.01562$), +-+ at $p=179$ ($p=0.04783$), and --+ at $p=179$ ($p=0.02879$). All other candidate ratios are below 1. The SHA control ratios across ascending p are 0.737, 0.683, 0.854, 0.794; the random-pair control ratios are 0.792, 0.680, 0.837, 0.785. The log-interval positive-control ratios are **2.595, 4.701, 8.549, 15.534**; each has empirical $p=1/2049$ and passes its 95th-percentile gate. The complete run has 64 observed spectra and **131,072 null spectra**.

The maximum inverse-FFT distance of a pair count from an integer is 7.19×10^{-9} ; maximum relative Parseval error is 1.46×10^{-12} . For inversion-symmetric sets, the forced $k+(-k)=0$ count dominates the maximum pair-sum deviation. That scalar is retained as a numerical check; it is not evidence of an exploitable relation. The full spectrum determines the ordered-pair sum vector and the earlier P+QES triple count through $(1/n) \sum_j F(j)^2 F(-j)$. The triple count itself was not separately calibrated. The maximum Fourier peak tests a different statistic from one triple count; no universal power ordering follows from this run.

Requirement status and interpretation limits

Requested element	Status
Predicate panel	Measured: five thresholds and eight signs; sizes and bitsets in JSON .
Null controls	Measured: SHA and random sets, 2048 matched nulls per cell.
Positive control	Measured: four of four log intervals pass.
FFT and convolution	Verified by direct DFT and pair-count tests.
Triple-count nulls	Not run; P+QES count is recoverable from bitsets.
Other proposed items	Items 2-6 not attempted here.

These observations are bounded to four toy curves and the specified predicate thresholds. The null ensemble conditions on set size and inversion symmetry; it does not prove that every algebraic or cheap predicate is flat. General Weil/Deligne estimates require a precisely specified sum and hypotheses, and the asserted hardcore-bit implication for arbitrary cheap-predicate/Fourier correlation needs a reduction with its own advantage and access assumptions. Neither claim is established by these measurements. A toy Fourier peak also does not imply a faster unknown-log algorithm. The user can treat the three unadjusted cells as data for future hypotheses; under this run's declared decision they are not leads.

Reproduction, validation, and repository graphs

From the repository root, with Rust 1.99 and the lockfile:

```
study=research/prime_fourier_flatness_exp7_20261007
cargo test --locked --release --example exp7_fourier_flatness
cargo run --locked --release --example exp7_fourier_flatness -- \
"$study/results_all_patterns.json"
cargo run --locked --release --example exp7_render_all_patterns -- \
"$study"
bash "$study/build_report_all_patterns.sh"
```

The measured JSON is **5,054,651 bytes**. Its SHA-256 is:

9322025e326761cf6e36b4fc67c85c12fe39300a3e365e005b7b6b78ee32de1c

Its source revision is recorded in the file. A same-command repeat written to /tmp/exp7-all-patterns-repeat.json was byte-identical (cmp exit 0; the same SHA-256). Four focused tests pass: FFT versus direct DFT and inverse, convolution versus direct pair counts, exact group enumeration, and predicate/null checks including all eight Legendre patterns. Targeted rustfmt --check and Clippy with the known unrelated clippy::mismatched_bit_width_type lint suppressed pass. Strict Clippy reports the existing src/symmetric/modes/pmac.rs:112 lint; repository-wide cargo fmt --all -- --check also reports pre-existing formatting elsewhere. Neither was caused by EXP7. The pilot report records its corrected premeasurement test assertion.

The sweep charged whole-group enumeration and all 131,072 null FFTs. It contains no timed-ratio, online DLP, or speedup claim. The canonical [index-calculus scoreboard](#), [progress data](#), docs/curves/, docs/performance-gains/, and figures/ were checked. EXP7 creates no IC/rho ratio, curve registry change, verified attack performance result, or curve edge, so those panels are unchanged. This study's new graph and diagram show the measured diagnostic and are kept with their editable sources and PDF copies.